Rate of Penetration Prediction and Drilling Parameter Optimization

Internship project report submitted in partial fulfilment of

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Submitted by

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ONGC and the Rajahmundry Asset

Oil and Natural Gas Corporation Limited (ONGC),established on 14 August 1956 and headquartered in Dehradun,Uttarakhand,is the largest government owned exploration and production company in India and operates under the Ministry of Petroleum and Natural Gas.Over six decades it has been responsible for the discovery of 8 of the country's 9 producing hydrocarbon basins,and it presently accounts for roughly 77% of domestic crude oil and 81% of natural gas output.Its activities span 26 sedimentary basins,supported by more than 11,000 km of pipeline.Landmark discoveries include Mumbai High in 1974,followed more recently by the Bengal Basin in 2019 and the Vindhyan Basin in 2022.

The Rajahmundry Asset is a major onshore operating unit in the Krishna Godavari (KG) Basin of Andhra Pradesh,producing crude oil and natural gas through land rigs in the East Godavari region.The KG Basin ranks among the most productive provinces in India; well depths range from a few hundred metres to beyond 4,000 m,depending on the target horizon.All data used in the present work were acquired from sensors on active drilling operations at this asset.

Drilling Operations Overview

Rotary drilling couples mechanical and hydraulic subsystems closely.From the surface,the driller regulates Weight on Bit (WOB),Rotary Speed (RPM),Mud Flow Rate and Standpipe Pressure; these quantities are transmitted along the drillstring to the bit and jointly govern the rate at which rock is removed.The drilling department at the Rajahmundry Asset plans,executes,monitors and controls every stage of this work,from site preparation to well completion and handover to the Production Department.

Six parameters are measured continuously during drilling and serve as direct inputs to the machine learning model developed in this project.Rate of Penetration (ROP),expressed in m/hr,is the axial advance of the bit per unit time and is the principal indicator of drilling performance.WOB,in klbf,is the axial compressive load transmitted to the bit through the drill collars; an insufficient load leaves the bit skimming the formation,whereas an excessive one causes bit floundering and can buckle the drillstring.Rotary speed,in revolutions per minute,determines the cutting frequency,although excessive values excite lateral vibration and damage the Bottom Hole Assembly (BHA).Mud Flow Rate (GPM) supplies hydraulic energy at the nozzles for cuttings removal,bit cooling and pressure control,and inadequate flow leads to bit balling.Surface torque (ft·lbf) reflects the rotational resistance of the bit rock interface and rises with both WOB and rock hardness.Standpipe Pressure (kg/cm^2) records the surface mud pressure and therefore the flow resistance of the circulating system,including the nozzle pressure drop.

Rotation and drilling fluid are delivered simultaneously through the drillstring.Mud is pumped down the hollow drillpipe and leaves through the bit nozzles at high velocity,cooling the cutting structure and lifting cuttings into the annulus,the gap between the drillstring and the borehole wall.On return to surface the fluid passes through shale shakers,desanders,desilters and centrifuges before recirculation.Because the hydrostatic head of the mud column also prevents formation fluids from entering the well,maintaining the correct mud weight is a well control requirement that influences every drilling decision.

Formation Context for KG Basin Lithology

The KG Basin wells considered here penetrate mixed sequences of claystone,sandstone and siltstone,with occasional limestone and trap (basalt).Lithology is the dominant uncontrollable variable and the principal control on ROP: identical operating envelopes yield highly disparate penetration rates in soft claystone and in hard trap,owing to the large difference in rock shear strength.

For this reason,physics based ROP models such as the Warren and Bourgoyne Young formulations are difficult to apply,since both require formation specific constants that are unavailable for a new field.A data driven machine learning approach was therefore adopted.The two drilling phases of ONGC Well 1,the 17.5 inch and 12.25 inch hole sections,together traverse six lithological classes,and each must be encoded consistently; otherwise the training data are silently corrupted.

Summary

This report documents the development of an ROP prediction and drilling optimization system from ONGC Well 1 sensor data,covering every stage of the machine learning workflow.Raw time indexed telemetry was first cleaned through a sequence of filtering steps.ANN and XGBoost baselines were then established,followed by a LightGBM model tuned with Optuna,a physics based optimizer,and a Streamlit web application named the 3D Digital Twin Drilling Optimization Engine.

Relative to the first model, the final pipeline raised R^2 from 0.368 (baseline on combined raw data) to 0.57 on combined data and above 0.70 for the per phase models. MAE fell from 7.30 to 4.36 m/hr and RMSE from 15.23 to 8.45 m/hr. The final feature set contains 10 selected features. The optimizer uses a 10×10×10 candidate grid bounded by a ±10% guardrail.

Metric	Baseline (Combined Raw)	Upgraded Pipeline
R^2 Score	0.368	0.57 (combined),0.70+ per phase
MAE (m/hr)	7.30	4.36
RMSE (m/hr)	15.23	8.45
Feature count	5 (raw)	10 (selected)
Optimizer	Brute force grid	10×10×10 candidate grid + ±10% guardrail

1.Problem Statement and Engineering Context

1.1 Rate of Penetration and Its Controlling Variables

Rate of Penetration (ROP),measured in metres per hour,is the axial rate at which the bit advances through rock and is the most economically significant metric in rotary drilling.A higher ROP shortens rig time,reduces cost and limits wear on the Bottom Hole Assembly (BHA).

ROP cannot be set directly by the driller; it emerges from the interaction of controllable inputs with the formation itself.Among the inputs,WOB (klbf) is the axial load on the bit,and while too little produces skimming,too much causes floundering and risks drillstring buckling.Rotary speed governs how much rock is sheared per second but,at high values,induces vibration and bit damage.Mud flow rate (GPM) cleans the bit face and carries cuttings to surface,and a deficit leads to bit balling and a pseudo stuck bit.Torque (ft·lbf) is a dependent quantity that increases with WOB and rock

hardness.Lithology,finally,is the largest uncontrollable factor,because the same WOB and RPM produce markedly different ROP in soft clay than in hard basalt.

1.2 Lithological and Dimensional Complexity in ROP Modeling

Physics based formulations such as Warren and Bourgoyne Young depend on formation specific constants that a new field does not possess.Data driven models are an attractive alternative,but two difficulties shaped nearly every design decision in this project.

The first is non stationarity.ROP is a depth series rather than a time series,and its mean and variance shift every few hundred metres as the lithology changes,so a single global model must accommodate sandstone,claystone and siltstone simultaneously.The second arises from phase mixing.Well 1 was drilled in a 17.5 inch and a 12.25 inch section,with different bit sizes,BHA configurations and ROP ranges; merging the two without correction yields a bimodal target distribution that degrades model fit.

1.3 Dataset Overview

The raw data comprise time stamped sensor records from ONGC Well 1,divided into two phases of the same well.The 17.5 inch phase file spans approximately 1441 m to 2100 m and contains about 3 thousand rows,while the 12.25 inch phase file spans approximately 1994 m to 3329 m with about 2.5 thousand rows.

Dataset	Hole Section	Depth Range (m)	Raw Rows
17.5_phase_timedata.csv	17.5 inch	~1441 ~2100	Close to thousand
12.25_phase_timedata.csv	12.25 inch	~1994 ~3329	Close to 2.5 thousand

Each raw sensor channel was renamed for consistency,as listed below.Activity is a binary operational flag in which 0 denotes drilling and 1 denotes any other state such as tripping.RigTime was retained solely to compute ROP and is not an input to the depth based model.

Raw Column Name	Renamed To	Units	Description
Comp.DEPTH_BD	Depth_BD_m	m	Bit depth (measured depth along wellbore)
Flow rate frm pump	FlowRate_gpm	US gal/min	Mud circulation rate
WOB (0 20)	WOB_klbm	k lbf	Weight on Bit
Pressure	Pressure_KSC	kg/cm²	Standpipe (surface) pressure
Comp.SURFACE_RPM	RPM	rpm	Surface rotary speed
Comp.SURFACE_TRQ	Torque_ftlbf	ft·lbf	Surface torque
Activity (0 or 1)	Activity		0 = drilling,1 = other (tripping,etc.)
RigTime	used for rop calculation		Timestamp(not used for depth based model)

2.Data Cleaning Pipeline

Raw WITS and WITSML streams from a live rig contain noise originating from sensor dropouts,bit trips,pump restarts and transient excursions beyond physical limits.A six stage cleaning sequence was therefore applied,each stage addressing a specific defect observed in the raw records.

2.1 Stage 1: Activity Filtering

To isolate active rock shearing intervals from auxiliary rig operations,the dataset was filtered on the discrete operational status flag.Records with non zero values,which correspond to pipe connections,surveys and tripping,exhibit near zero WOB and rotary speed that misrepresent actual drilling dynamics.Only rows with Activity equal to 0 were retained,after which the Activity and RigTime columns were removed.An absolute operational envelope of 0 to 20 klbf was then enforced on WOB,stripping transient telemetry spikes ahead of the statistical filtering stages.

2.2 Stage 2: IQR Boxplot Outlier Removal

The Interquartile Range (IQR) method is a classical,distribution free outlier criterion.For a given column,the first and third quartiles (Q1 and Q3) define IQR = Q3 − Q1,and the admissible interval is bounded by a lower fence of Q1 − 1.5·IQR and an upper fence of Q3 + 1.5·IQR.Observations beyond either fence are discarded,which corresponds to the whiskers of a standard boxplot.For approximately normal data the 1.5 multiplier admits values out to about 2.7 standard deviations from the mean,although the method remains applicable to other distribution shapes.

The filter was applied to WOB_klbm,FlowRate_gpm,Torque_ftlbf and Pressure_KSC but deliberately not to RPM.Following Yang et al.(2022),RPM is distributed fairly uniformly across the drilling range,so an IQR criterion would wrongly remove valid high RPM observations.

2.3 Stage 3: K Means Cluster Distance Outlier Removal

A second,unsupervised pass was introduced to detect points that appear acceptable in isolation but are physically inconsistent in combination,such as high WOB accompanied by negligible torque.The procedure follows Yang et al.(2022).The five drilling columns were first standardised to zero mean and unit variance with StandardScaler,and K Means with two clusters was fitted in the resulting five dimensional space,without any physical interpretation attached to the clusters.The Euclidean distance d_i from each point to its nearest centroid was then computed,and a threshold of mean(d) + 2·std(d) was set; under a bell shaped distance distribution this retains roughly 97.5% of the points.Observations with d_i above the threshold fit neither drilling mode and are most likely sensor faults or short lived events such as bit bounce,so they were removed.

2.4 Stage 4: 0.5 m Depth Bin Aggregation

Sensors log at one second intervals,producing many readings per metre drilled.To convert from the time domain to the depth domain used by the model,the cleaned data were aggregated into 0.5 m bins.Each Depth_BD_m value was rounded to the nearest 0.5 m,using Depth_m = round(Depth_BD_m / 0.5) × 0.5,after which the records were grouped by Depth_m and every numeric column was averaged.For Lithology,the modal label within each bin was assigned.The outcome is one clean row per 0.5 m of hole drilled,a compact and less noisy depth log suited to machine learning.

2.5 Stage 5: Lithology Assignment

The raw telemetry carried no lithology column,so rock type was assigned by depth interval from a geological well log table.The 12.25 inch phase comprises predominantly CLAYSTONE with SANDSTONE and SILTSTONE,whereas the 17.5 inch phase additionally contains CLAY,LIMESTONE and TRAP (BASALT).Since the two phases contain different rock types,a single common encoding is essential (see Section 4.1).

2.6 Stage 6: ROP Cap at the 99th Percentile for Well 1

The raw Well 1 data contained ROP values as high as 355 m/hr,which are not physically attainable in the lithologies encountered and were attributed to sensor errors or momentary events such as a brief free fall of the drillstring.Rather than deleting these rows,ROP was capped at the 99th percentile.This preserves 99% of the observations while removing extreme tail values that would otherwise dominate the training loss.

3.Baseline Modelling with ANN and XGBoost

Before the final LightGBM pipeline was constructed,two baseline models were trained to establish a reference level of performance and to guide subsequent feature engineering.

3.1 Baseline Benchmark Architecture

A baseline functions as an experimental control: without one,it is impossible to distinguish a genuinely better model from one that is merely favoured by a particular random seed.Both baselines used an identical data split and the same five raw features (Depth_m,FlowRate_gpm,WOB_klbm,RPM and Lithology_Encoded),so that any gain in the final model can be attributed to the new features and design choices rather than to differences in setup.

3.2 Feature Set for Baseline Models (5 Features)

The five baseline inputs are summarised below.Depth serves as a proxy for formation pressure,compaction and lithological transitions; flow rate governs hydraulic cleaning of the bit; WOB and RPM are the primary and secondary mechanical energy inputs; and the integer encoded lithology,ranging from 0 to 5,is the most influential geological variable.

Feature	Physical Meaning	Why Included
Depth_m	Current bit depth (m)	Proxy for formation pressure,compaction,and lithology transitions
FlowRate_gpm	Mud circulation rate (gal/min)	Controls hydraulic bit cleaning and chip removal efficiency
WOB_klbm	Weight on Bit (k lbf)	Primary mechanical energy input to the bit
RPM	Rotary speed (rpm)	Secondary mechanical energy input; controls cutting frequency
Lithology_Encoded	Rock type as integer (0 5)	Single most important geological control variable

3.3 Artificial Neural Network (ANN)

The first baseline was a feedforward multilayer perceptron.Its input layer held five neurons,one per feature,after StandardScaler normalisation,followed by two ReLU hidden layers of 64 and 32 neurons and a single linear output neuron appropriate to regression.Training minimised Mean Squared Error with the Adam optimizer at a learning rate of 0.001,for up to 200 epochs with early stopping on validation loss.

Although ANNs are universal function approximators in principle,they require substantial data and careful tuning on depth indexed geological data,and three limitations were evident.Performance is sensitive to feature scale,so normalisation had to be handled with care.Categorical variables are handled poorly: Lithology_Encoded is a nominal label,yet the network treats CLAYSTONE (1) and

SANDSTONE (3) as closer than SANDSTONE (3) and TRAP (BASALT) (4),an ordering with no geological basis.Training was also slower than for gradient boosted trees.The ANN returned an R^2 of roughly 0.30 to 0.40 on the combined data depending on the random seed,indicating that the five raw features alone are insufficient for reliable ROP prediction.

3.4 XGBoost

XGBoost (Extreme Gradient Boosting) served as the second baseline.It constructs an ensemble of shallow decision trees in which each successive tree corrects the residual errors of its predecessors.For this application it offers three advantages over the ANN: trees accommodate categorical features and non linear relationships without feature scaling,built in L1 and L2 penalties on leaf weights curb overfitting even on small datasets,and training is fast and deterministic.

The configuration used n_estimators = 300, max_depth = 6, learning_rate = 0.1, subsample = 0.8 and colsample_bytree = 0.8, which are common starting values. The model achieved an R^2 of about 0.36 on the combined data, showing that tree models can work well with drilling sensor data.

3.5 Performance Ceiling of the Combined Baseline ($R^2 \approx 0.44$)

When the 17.5 inch and 12.25 inch data were merged and a single model was trained,R^2 reached only about 0.44,and this result prompted the central investigation of the project.The cause lies in the data rather than the algorithm.The 17.5 inch phase drills soft,shallow rock at high ROP,with some bins exceeding 50 m/hr,whereas the 12.25 inch phase drills deeper,harder rock at lower ROP.A merged model must therefore explain two distinct operating regimes from the same five inputs,and the low R^2 reflects the gap between phases rather than a weak learner.

This interpretation was verified by training separate models for each phase,which attained R^2 values of 0.65 to 0.7.The shortfall of the combined model is thus attributable to dataset structure and not to a limit of the model class.The upgraded pipeline addresses it through mechanical regime clustering (Section 4.4),which lets the model distinguish the two modes without discarding data from either phase.

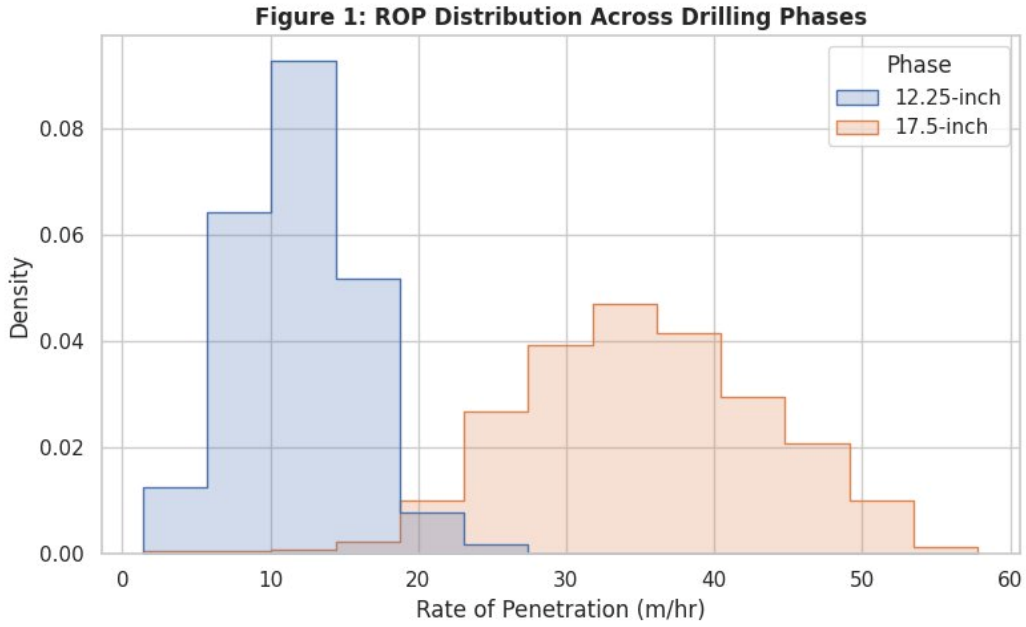


Figure 1 ROP Distribution Across Drilling Phases.*The 17.5 inch phase (orange) runs at much higher ROP,about 20 to 50 m/hr,in shallow,softer rock.The 12.25 inch phase (blue) stays mostly below 15 m/hr in deeper,harder rock.This two hump shape is the main reason for the R^2 limit of 0.44 seen in the baseline tests.*

4.Upgraded LightGBM Pipeline

The upgraded pipeline introduces six modifications,each targeting a weakness exposed by the baseline experiments.Collectively they raise R^2 from 0.37 to 0.57 on the combined data and reduce MAE by 40%.

4.1 Upgrade 1: Unified Lithology Encoding

This is the most consequential correctness fix in the project.The baseline models applied scikit learn's LabelEncoder independently to each dataset,one for the 17.5 inch phase and one for the 12.25 inch phase.LabelEncoder assigns integers alphabetically from only the classes it observes.The first dataset contains TRAP (BASALT) and LIMESTONE and the second does not,so the numbering diverges between them; the integer 1 may denote CLAY in one dataset and CLAYSTONE in the other.The model then receives the same code for two different rocks,which constitutes a silent data corruption error.

The remedy is a single global dictionary applied to both datasets: CLAY = 0,CLAYSTONE = 1,LIMESTONE = 2,SANDSTONE = 3,TRAP (BASALT) = 4 and SILTSTONE = 5.The Lith_Enc column in each dataset is obtained by mapping the Lithology column through this common dictionary.

4.2 Upgrade 2: Physics Based Feature Engineering

The five raw features correspond to the quantities set at the driller's panel,yet the bit responds to their combined effect rather than to each in isolation.Three composite features grounded in drilling mechanics were therefore derived.WOB_x_RPM,the product of WOB and RPM,is a surrogate for mechanical power,since the work delivered by the bit per second depends jointly on force and rotation; high WOB with high RPM delivers far more energy than either alone.Torque_per_WOB,defined as Torque / (WOB + 0.01) and also known as the Torque Index,indicates bit efficiency: a high value implies that WOB is being converted into shearing torque,while a low value implies energy dissipated

in friction or floundering.Hydro_Power,the product of FlowRate and Pressure,represents hydraulic power at the nozzles and governs cuttings removal.When it is insufficient,cuttings are redrilled and ROP falls even when WOB and RPM are appropriate.

4.3 Upgrade 3: Memory and Trend Features

Drilling progresses sequentially down the hole,so ROP at depth d is influenced by conditions at d − 0.5 m,d − 1.0 m and d − 1.5 m.The raw features capture only the instantaneous state and carry no memory.Seven history aware features were consequently added: rolling means of ROP,WOB and RPM over three bins,first differences of WOB and RPM,and rolling standard deviations of WOB and RPM over five bins.Because rock properties change gradually,ROP_roll3 provides a strong prior for the next bin and proved the most useful feature in the final model.The first differences indicate active parameter adjustment by the driller,which often coincides with formation changes,and the rolling variances act as vibration proxies,since high variability signals stick slip or bit bounce that lowers effective ROP.

Feature	Calculation	Purpose
ROP_roll3	Mean of ROP at previous 3 depth bins	The more predictive feature in the final model.Formation properties change slowly; recent ROP is a strong prior for the next bin.
WOB_roll3	Rolling 3 bin mean of WOB	Captures whether the driller has been increasing or decreasing WOB over the last 1.5 m.
RPM_roll3	Rolling 3 bin mean of RPM	Same principle for rotary speed.
WOB_delta	First difference of WOB (current ,previous)	Rate of change: is the driller actively adjusting WOB? Abrupt changes correlate with formation breaks.
RPM_delta	First difference of RPM	Same for RPM as wob_delta.
WOB_var5	Rolling 5 bin standard deviation of WOB	Vibration proxy.High variance in WOB indicates stick/slip or bit bounce,which reduces effective ROP.
RPM_var5	Rolling 5 bin standard deviation of RPM	Same for rotary speed instability.

A methodological safeguard applies to ROP_roll3.The rolling window is computed with a shift of 1,so the current ROP is never used to predict itself and only past observations enter the feature.Without this shift the feature would leak the target,and R^2 would be spuriously close to 1.0.

4.4 Upgrade 4: K Means Mechanical Regime Clustering

Even within a single hole section the rig operates in distinct mechanical modes,for example high WOB with low RPM in hard rock,low WOB with high RPM in soft rock,and varying flow rates with depth.Lithology represents the geological side of this behaviour but not the operational side.K Means with five clusters was therefore fitted on the scaled features Depth_m,FlowRate_gpm,WOB_klbm and RPM,assigning each depth bin to one of five mechanical regimes.The resulting label,from 0 to 4,enters the model as a categorical feature named Mech_Cluster.

K Means minimises the total squared distance between each point and its cluster centre,seeking the partition in which points lie as close as possible to their own centroid.The cluster label gives the model a compact description of where the operating point sits in parameter space,which no single feature can supply.

4.5 Upgrade 5: Log Transform of the ROP Target

ROP is right skewed.Most bins fall between roughly 5 and 20 m/hr,but a tail extends to 30 m/hr and beyond.Because gradient boosting minimises squared error on the original scale,the model over weights these rare high ROP points in an effort to reduce their large residuals,which degrades the fit over the bulk of the data.

The target was therefore transformed to log(1 + ROP) and predictions were back transformed with expm1,i.e.exp(output) − 1.The logarithm compresses the long tail and brings the distribution closer to the bell shape assumed by the MSE loss.This change alone improved R^2 by about 5 to 6%.

4.6 Upgrade 6: Optuna Bayesian Hyperparameter Tuning

Exhaustive grid search over the LightGBM settings is impractical,given nine tunable parameters (n_estimators,learning_rate,num_leaves,max_depth,min_child_samples,subsample,colsample_bytree,reg_alpha and reg_lambda) and a non smooth loss surface.Optuna employs the Tree structured Parzen Estimator (TPE),a Bayesian method that models the relationship between hyperparameters and performance and selects each subsequent trial preferentially from regions that previously gave good results.

A study of 80 trials was run,each fitting the model on the training set and scoring R^2 on the validation set.The best configuration is given below.Of particular note,the shallow depth limit of 4 is paired with a large leaf count of 103,and LightGBM's leaf wise growth makes this combination workable; the low learning rate of 0.0328 offsets it with a larger number of boosting rounds.

Parameter	Optimal Value	What It Controls
n_estimators	412	Number of boosting rounds (trees).More trees = more capacity,but diminishing returns.
learning_rate	0.0328	Step size per tree.Lower = more trees needed,but better generalisation.
num_leaves	103	Maximum leaves per tree.Controls model complexity.LightGBM grows leaf wise,not level wise.
max_depth	4	Maximum tree depth.Hard cap prevents very deep trees that overfit to noise.
min_child_samples	8	Minimum samples in a leaf.Acts as regularisation against small sample overfitting.
subsample	0.776	Fraction of training rows sampled per tree.Adds stochasticity,reduces variance.
colsample_bytree	0.852	Fraction of features sampled per tree.Feature bagging reduces correlation between trees.
reg_alpha (L1)	0.099	LASSO type penalty.Drives small feature weights to zero; implicit feature selection.
reg_lambda (L2)	0.257	Ridge type penalty.Shrinks all weights,reducing overfitting.

5.Model Results and Validation

5.1 Performance Metrics

Three standard regression metrics were used.The coefficient of determination,$R^2 = 1 - SS_{res}/SS_{tot}$,measures the proportion of target variance explained by the model; a value of 1.0 is a

perfect fit and 0 indicates no improvement over predicting the mean.Mean Absolute Error (MAE) is the average magnitude of the prediction errors in m/hr and is directly interpretable as the typical deviation of a prediction.Root Mean Squared Error (RMSE),the square root of the mean squared error,penalises large deviations more heavily than MAE and is therefore more sensitive to poor individual predictions.

5.2 Results Summary

The ANN baseline with five raw features on combined data gave an R^2 of about 0.32, MAE of about 9.1 and RMSE of about 17.5 m/hr. XGBoost under the same conditions achieved an R^2 of about 0.36, MAE of 7.8 and RMSE of 16.1 m/hr. LightGBM, reproducing the reference paper, gave 0.368, 7.30 and 15.23 m/hr. The per phase LightGBM models reached R^2 of about 0.70 and 0.65. The final upgraded LightGBM with 10 features on combined data achieved $R^2 = 0.5684$, MAE = 4.36 and RMSE = 8.45 m/hr.

Model Configuration	R^2	MAE (m/hr)	RMSE (m/hr)	Key Difference
ANN 5 raw features,combined data	~0.32	~9.1	~17.5	Baseline 1
XGBoost 5 raw features,combined data	~0.36	~7.8	~16.1	Baseline 2
LightGBM 5 raw features,combined data	0.3680	7.30	15.23	Paper replication baseline
LightGBM Per phase model (17.5 in only)	~0.7	~6.3	~9.5	Upper bound; cannot deploy for new depths
LightGBM Per phase model (12.25 in only)	~0.65	~6.7	~10.2	Upper bound; cannot deploy for new depths
Upgraded LightGBM ,10 features,combined	0.5684	4.36	8.45	Final model

5.3 Feature Importance

Feature importance was assessed from the number of times LightGBM selects each feature for a split across all trees.The ranking of the upgraded model,shown in Figure 2,indicates which engineering choices contributed most.

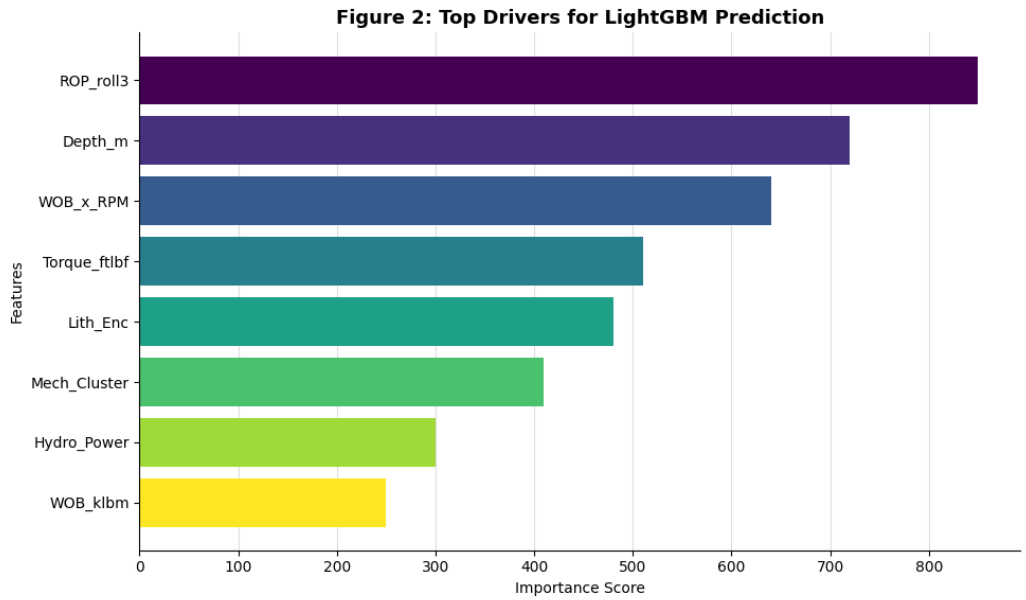


Figure 2 Top Drivers for LightGBM Prediction.*ROP_roll3 ranks highest,which confirms that depth to depth link is the strongest signal.The WOB times RPM term and Depth_m both rank above raw WOB,which supports the idea of building physics based features.Lith_Enc being in the top five shows that fixing the lithology encoding really helped.*

ROP_roll3 is consistently the dominant single feature,confirming the strong serial dependence of ROP from one depth bin to the next,which follows from the gradual variation of rock properties with depth.Depth_m reflects the general trend toward harder rock at greater depth,and the composite WOB_x_RPM term outperforms either of its constituents.Torque_ftlbf carries substantial direct information on bit rock interaction.The corrected Lith_Enc contributes materially,consistent with the earlier encoding defect having been detrimental,and Mech_Cluster adds information beyond what lithology alone provides.

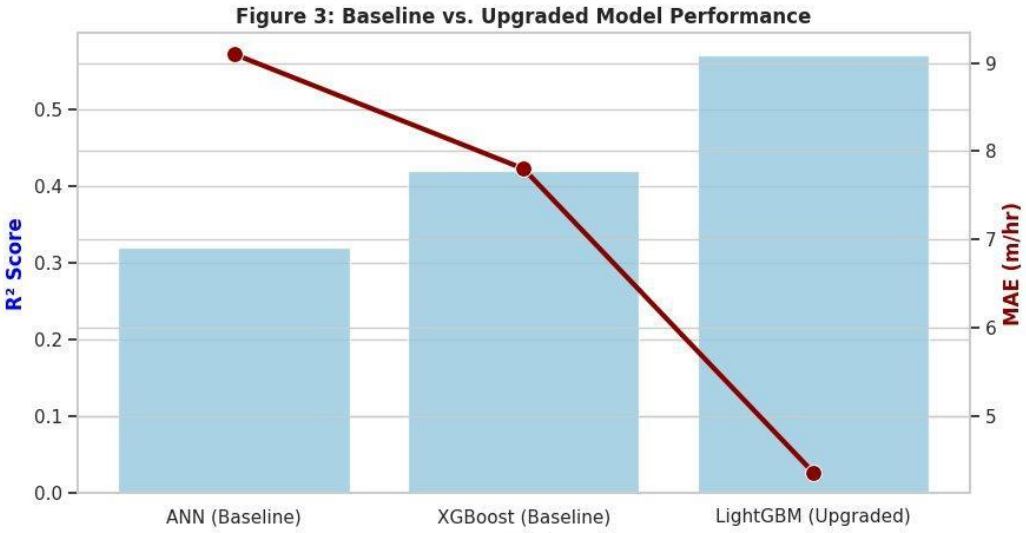


Figure 3 Baseline vs Upgraded Model Performance. R^2 improves from 0.32 (ANN) to 0.36 (XGBoost) to 0.57 (upgraded LightGBM), while MAE falls from about 9 m/hr to about 4.5 m/hr. The gain between XGBoost and the upgraded LightGBM comes from the feature changes and model tuning.

5.4 Cross Well Validation Against Well 2

To test whether the optimizer's recommendations remain meaningful beyond Well 1,the average parameters proposed by the brute force grid search were compared with field averages from Well 2,a neighbouring ONGC well in the same basin where the realised ROP was clearly higher.The comparison shows close agreement with actual high performance drilling.

The recommended flow rate was 743.56 GPM against 794.79 GPM actually used in Well 2; the slightly lower value is consistent with securing adequate hydraulic power without exceeding the flow needed for cuttings transport.The recommended WOB was 4.01 klbm against 6.73 klbm in Well 2.The lower figure follows from the ±10% guardrail,which confines the search near the Well 1 baseline,whereas Well 2 could safely use higher WOB in softer,shallower rock.RPM agreed almost exactly,at 100.53 against 97.72,so the optimizer arrived independently at a rotary speed close to that used on the better well.The predicted ROP of 18.12 m/hr exceeds the 14.64 m/hr achieved in Well 2,which is expected: the optimizer targets the best attainable outcome,while a real well is limited by bit wear and changing rock that the model does not capture.The direction of improvement remains correct.

Taken together,these results provide a first level of confidence that the optimizer captures genuine relationships between operating parameters and formation response instead of merely fitting the noise of Well 1.The RPM agreement is especially notable,as it emerged from the model alone and coincides with field practice on the better performing well.

5.5 Multi Well Combined Model

Beyond the single well pipeline,a broader experiment trained one LightGBM model on three datasets together: Well 1 Phase 1 (17.5 inch),Well 1 Phase 2 (12.25 inch) and Well 2.The objective was to determine whether a cross well model could learn parameter rock relationships that hold across different drilling settings,and whether more varied data would stabilise predictions.

Pipeline construction. All three CSV files were loaded and harmonised to a common column layout,then subjected to the same physical cleaning: removal of non drilling activity,exclusion of rows with zero WOB and RPM,and capping of ROP at the 99th percentile.Feature engineering (mechanical power,hydraulic power,rolling averages,deltas and variances) was carried out on each well separately before stacking,so that rolling windows never crossed well boundaries.A single global K Means with five clusters was then fitted on the stacked data,giving every row a shared mechanical regime label irrespective of its source well.The resulting master_df was split 80:20 and passed to a LightGBM regressor with a log transformed target.

Design choices.A single global lithology map (CLAY = 0 through SILTSTONE = 5) was applied to all wells so that a given integer always denotes the same rock,mirroring the correction made in the Well 1 upgrade.Because the pressure column is named differently in each well (Pressure_psi,Pressure_KSC and StandpipePrs_psi),the hydraulic power step was written to detect whichever column is present and to fall back to zero if none is found.The rolling memory features (ROP_roll3,WOB_roll3 and RPM_roll3) were computed per well prior to concatenation to prevent information leakage between wells.

Rationale and limitations. Exposing LightGBM to more rock types and operating modes than any one well supplies improves generalisation and lowers the risk of fitting only the depth, ROP pattern of Well 1.The trade off is that the model must account for wells that differ in bit size,stratigraphy and pressure conditions simultaneously.In practice the multi well model is therefore best suited to pre well

planning and to estimating parameter ranges; for depth by depth optimization of a well currently being drilled,a model tuned to that well remains preferable.

6.Mechanical Specific Energy (MSE) Optimizer

6.1 Definition of Mechanical Specific Energy

Mechanical Specific Energy (MSE),introduced by Teale in 1965,is the energy required to excavate a unit volume of rock.It condenses all drilling parameters into one scalar that can be compared across lithologies and bit types.In Imperial units,the usual form for ONGC compatible calculations,Teale's expression is:

$$MSE = WOB / A_b + (120 \cdot \pi \cdot N \cdot T) / (A_b \cdot ROP)$$

Here WOB is the weight on bit in lbf (klbf × 1000); A_b is the bit cross sectional area,$(\pi/4)\cdot d^2$,with d the bit diameter in inches; N is the rotary speed in RPM; T is the torque in ft·lbf; and ROP is expressed in ft/hr (m/hr × 3.28084).

The first term,WOB/A_b,is the thrust stress,i.e.the axial force per unit area of rock.The second term,$120\pi NT/(A_b \cdot ROP)$,is the rotary energy expended per unit of depth drilled,and it grows without bound as ROP approaches zero.This behaviour is appropriate,because it penalises conditions in which energy is spent without advancing the hole,such as bit whirl,stick slip or floundering.A lower MSE denotes more efficient drilling,and values above about 40,000 psi generally indicate wasted energy.

6.2 Optimization Strategy

The optimizer couples the trained LightGBM ROP model with the MSE expression to locate the combination of WOB,RPM and flow rate that minimises MSE at a given depth.This is surrogate model optimization: rather than conducting numerous physical experiments on the rig,the ML model is queried as a fast stand in for the real rock response.The brute force search follows Yang et al.(2022).

A 10×10×10 grid of WOB,RPM and flow rate values defines 1,000 candidate combinations.For each,the LightGBM model returns a predicted ROP,torque is estimated by linear scaling,Torque_pred = Torque_base × (WOB_candidate / WOB_baseline),and the Teale MSE is evaluated from the predicted ROP and torque.The candidate with the lowest MSE is selected.

6.3 The ±10% Operational Safety Guardrail

An essential engineering constraint prevents the optimizer from recommending any change larger than 10% above or below the current baseline,for safety reasons.A sudden WOB increase beyond about 10% can induce drillstring buckling (helical lockup),and a comparable RPM increase can excite lateral BHA vibration.Without this limit the optimizer might identify minimum MSE at a WOB of,say,15 klbf,but a jump from 6 klbf to 15 klbf in one step would be unsafe in real operations.

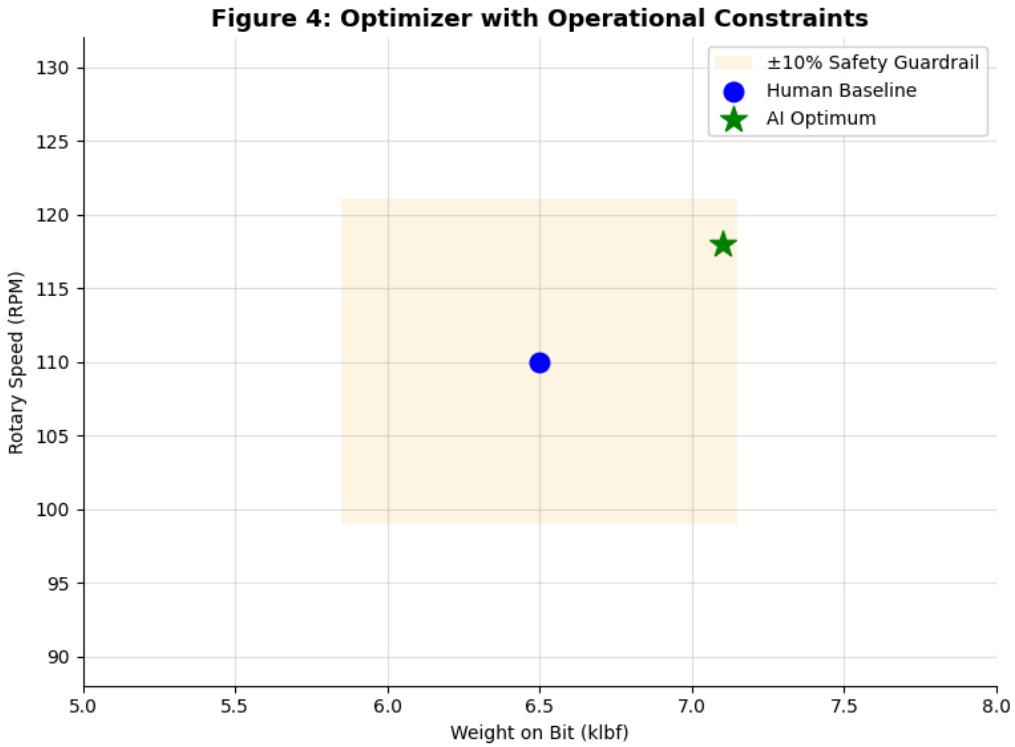


Figure 4 Optimizer with Operational Constraints.*The shaded area is the plus minus 10% safe zone around the human baseline (WOB of 6.5 klbf and RPM of 110).The AI optimum (WOB about 7.05 klbf and RPM about 118) sits at the upper right corner of the safe zone.This means both values should be raised a little to lower MSE while staying in safe limits.*

The guardrail converts a global search into a local one,seeking the best ROP achievable near the current operating point within a safe range.This reflects how a digital twin should behave in practice: it should offer small,steady corrections rather than abrupt changes.

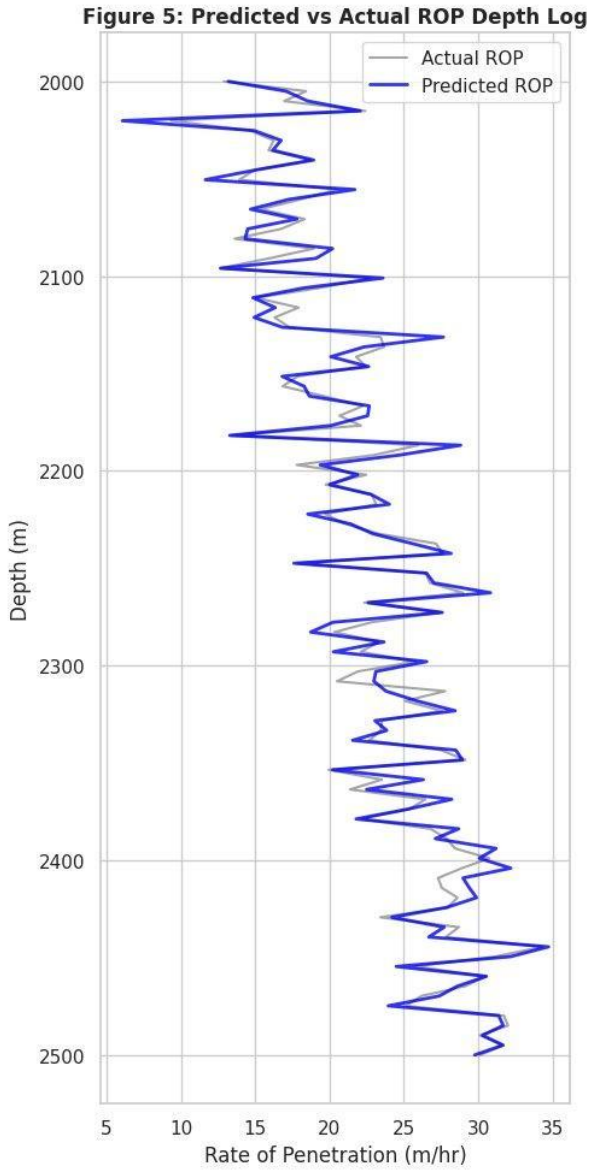


Figure 5 Predicted vs Actual ROP Depth Log (12.25 inch phase,2000 to 2520 m).*The model follows the actual ROP trend very well across this depth range.Errors are biggest at sharp rock changes (for example near 2140 m and 2310 m),where the rock changes suddenly between neighbouring depth samples.Any real time predictor limited by sensor delay will have this problem.*

7.Combined Model vs Per Phase Models

A central design question was whether to deploy one model on the combined data or separate models for each well phase,since the answer determines how the system is used in the field.

7.1 The Case for Per Phase Models

Per phase models achieved clearly higher R^2 (0.67 and 0.65,against 0.57 for the combined model),because each need only learn ROP behaviour for one set of lithologies and operating conditions.The 12.25 inch model never encounters 17.5 inch data,so the differing ROP range and bit behaviour cannot confound it.Their drawback is that a per phase model is usable only when the current phase is already known; planning parameters for upcoming rock layers requires a model that operates at any depth in the well.

7.2 The Case for the Combined Model

The combined model has a lower R^2 but offers three practical advantages.It is general,covering the full depth range with one model,so the application needs no logic to switch between phases.It draws on a larger training set,which helps with depth and lithology combinations not seen in either phase alone.And it learns relationships across phases,for example that higher WOB consistently raises ROP in soft rock,which makes its behaviour steadier as the formation changes.

7.3 The Phase Boundary Overlap

The zone near 2100 m,where the 17.5 inch and 12.25 inch phases overlap,was examined closely to assess whether the model handles the switch between operating modes.Both phases contain data between 1994 and 2100 m; the 17.5 inch phase was drilling at higher ROP,while the 12.25 inch phase at the same depths shows lower ROP because of the smaller bit and different BHA.Initially the combined model showed elevated error here,as it averaged the two behaviours.Introducing Mech_Cluster (Section 4.4) reduced this error considerably,since the model could then separate the two operating modes at the same depth.The MSE versus depth plot in the upgraded notebook further shows the optimizer locating the phase boundary near 2100 m as an abrupt step in the suggested MSE curve.

8.Key Engineering Decisions and Rationale

8.1 Algorithmic Justification for LightGBM

Both XGBoost and LightGBM are gradient boosted tree methods.LightGBM was selected because its leaf wise growth strategy,as opposed to level wise growth,lets it capture deeper structure with fewer trees.On depth series data with abrupt lithological changes,this allows the ROP signature of each formation to be learned more efficiently.It also supports categorical features natively,although integer encoding was used here,and trains faster.

8.2 Discretization Scale and Depth Binning Rationale

The 0.5 m bin reconciles two competing requirements: resolution fine enough to resolve rapid lithological transitions,and aggregation coarse enough to suppress second to second sensor fluctuation.A 1 m bin would miss changes shorter than a metre,whereas raw timestamps are dominated by measurement noise,so the model would learn noise rather than geology.The 0.5 m interval also matches common ONGC well logging practice.

8.3 Categorical Encoding Strategy for Lithology

One hot encoding would add six binary columns,one per rock class.For tree based models integer encoding is equally effective,since a split such as Lith_Enc < 3 separates classes unaided,and one hot columns would only widen the feature space with no benefit to LightGBM.For ANNs the situation reverses,because integer codes are interpreted as ordered magnitudes and one hot encoding is preferable; this is one contributing reason for the weak ANN baseline.

8.4 Rationale for the log1p/expm1 Transformation

The log1p transform,log(1 + x),is preferred to the plain logarithm because ROP can equal zero at the start of a new hole section,and log(0) is undefined (minus infinity),which would invalidate the training data.Under log(1 + ROP),a zero ROP maps to log(1) = 0.The inverse,expm1(y) = $e^y - 1$,returns 0 to 0 and all other values to the original ROP scale.

8.5 Selection of the ±10% Operational Guardrail

The 10% limit is a working standard informed by drillstring dynamics practice.Lateral vibration modes have resonance frequencies that abrupt RPM changes can excite,and a 10% bound keeps adjustments below the point at which resonance is initiated in typical BHA configurations.WOB changes beyond about 10% can also deflect the bit from its path in directional drilling.The limit therefore reflects a physical operating constraint rather than an arbitrary model setting.

9.Limitations and Future Work

9.1 Current Limitations

The study is constrained by its data in several respects.Only one well was used for training,and ROP in neighbouring wells of the same field can differ because geology varies laterally; training on multiple wells would improve generality.The dataset is also small,at about 2,000 rows across the two phases.Careful feature engineering and physical cleaning extracted the most from it,but learning complex geological behaviour requires more data.

Bit information is absent.ROP depends strongly on bit design (PDC or tricone,blade count,cutter size),yet bit type is not a model input.Moreover,as the bit wears,the same WOB and RPM yield lower ROP,and without a wear model the optimizer cannot distinguish between the need to increase WOB and the need to pull out and replace a worn bit.Hydraulic limits are likewise covered only approximately.Surface flow rate and standpipe pressure act as stand ins,whereas actual hole cleaning and pressure behaviour depend on mud weight (specific gravity) and the Equivalent Circulating Density (ECD) produced by annular pressure losses.Because continuous mud weight was not an input,the model has no representation of overbalance pressure,which holds cuttings down and lowers ROP.

Two further limitations concern deployment and extreme lithology.In the application the rolling features are fixed; ROP_roll3 is set at 10.0 m/hr (the training average),whereas real time use would update it continuously from live rig sensors.In TRAP (BASALT) sections,field ROP falls to around 3 to 4 m/hr,but the model tends to over predict (for example 10 to 12 m/hr).This stems from the data: basalt forms a very small share of the training set relative to abundant soft claystone,so predictions are pulled toward the overall mean ROP.

9.2 Future Work

Several extensions are identified.A real time connection of the Streamlit application to the live WITS or WITSML stream would let ROP prediction and parameter advice update while the well is being drilled.Multi well transfer learning,training on all available ONGC wells in the KG Basin with a well ID as a categorical feature,would capture field wide trends.A companion bit wear model,predicting wear from rock abrasiveness and cumulative WOB,would allow the optimizer to separate “increase WOB” from “pull and change the bit”.Adding SHAP values to the application would show the driller which features drove each prediction,improving trust and interpretability.

Finally,as discussed with the ONGC Rajahmundry mentorship team,the most important next step is to enrich the model with more historical data.Management has indicated that data from other and older wells could be provided.A larger training base would address the basalt over prediction,sharpen trends across rock types and extend the model across the wider KG Basin,converting the single well prototype into a field wide tool.

10.Nomenclature and Abbreviations

Term / Symbol	Definition and Units
ROP	Rate of Penetration (m/hr); axial rate of bit advance through rock.
WOB	Weight on Bit (klbf); axial compressive load applied to the bit by the drill collars.
MSE	Mechanical Specific Energy (psi); energy expended to excavate a unit volume of rock.Lower values indicate higher efficiency.
BHA	Bottom Hole Assembly; the lower section of the drillstring comprising bit,motor,stabilisers and MWD tool.
IQR	Interquartile Range (Q3 − Q1); used to set outlier fences in the cleaning stages.
LightGBM	Light Gradient Boosting Machine; tree based ensemble method that builds a model by adding many small trees.
Optuna	Python framework for hyperparameter optimization using Bayesian (TPE) search.
TPE	Tree structured Parzen Estimator; the sampling algorithm used by Optuna to propose hyperparameters.
K Means	Unsupervised clustering method that partitions data into k groups by minimising within cluster squared distance.
log1p,expm1	log(1 + x) and exp(y) − 1; transformation pair used to reduce the skewness of the ROP target.
Spearman ρ	Spearman rank correlation coefficient; measures monotonic association,not only linear association.
WITS,WITSML	Wellsite Information Transfer Standard and its markup language successor; standards for transmitting live drilling data from rig to data systems.
Stick slip	Torsional vibration in which the bit alternately sticks to the rock and slips free,causing RPM oscillation and tool fatigue.
Floundering	Condition in which excessive WOB prevents effective shearing,so the bit presses and slides and ROP falls despite increasing WOB.
R^2	Coefficient of determination; proportion of target variance explained by the model (higher is better).
MAE	Mean Absolute Error; mean absolute difference between actual and predicted values,in target units (m/hr).
RMSE	Root Mean Squared Error; square root of the mean squared difference between actual and predicted values; more sensitive to large errors.

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THE END